Randomized and low-rank approximation

Clustered singular-value gaps in randomized block Krylov approximation

Difficulty: hard **Importance:** interesting to specialist

Status: source-stated conjecture; no resolution found in screening on 2026-09-08.

Let $A \in \mathbb{R}^{n\times d}$ have singular values $\sigma_1 \geq \sigma_2 \geq \cdots$. Fix integers $1 \leq b \leq k$, set $t = \lceil k/b \rceil$, and set $k' = bt \leq \operatorname{rank}(A)$. Define

$$\Delta_{k'}^{(b)} = \min_{1\leq i\leq k'-b} \frac{\sigma_i^2 - \sigma_{i+b}^2}{\sigma_i^2},$$

with the empty minimum set to one, and suppose this gap is positive. Draw $G \in \mathbb{R}^{n\times b}$ with independent standard Gaussian entries. In exact arithmetic, let Z be an orthonormal basis of

$$\operatorname{range}[G, (AA^T)G, \ldots, (AA^T)^{q-1}G], \qquad \widehat{A} = Z[Z^T A]_k,$$

where $[B]_k$ denotes a best rank-k approximation obtained by truncating the SVD of B.

Does an absolute constant C exist such that, for every such input and every $0 < \varepsilon, \delta < 1/2$, taking

$$q = \left\lceil C \left[\frac{t}{\sqrt{\varepsilon}} \log \frac{2}{\Delta_{k'}^{(b)}} + \frac{1}{\sqrt{\varepsilon}} \log \frac{n}{\delta\varepsilon} \right] \right\rceil$$

gives, with probability at least $1-\delta$, both

$$\|A - \widehat{A}\|_\xi \leq (1+\varepsilon)\|A - [A]_k\|_\xi \quad (\xi = 2, F),$$

and $|\|Av_i\|_2^2 - \sigma_i^2| \leq \varepsilon\sigma_{k+1}^2$ for the ordered top k right singular vectors v_i of $\widehat{A}$? Here $\sigma_{k+1} = 0$ if necessary. The factor two makes the logarithm meaningful at a unit gap.

This is the gap-independent algorithmic component of the authors' concluding conjecture. They also conjecture the corresponding improvement in their gap-dependent theorem and random Krylov matrix conditioning theorem; those are not separately counted here. The displayed parameter regime is where the source's spectral-gap expression is finite and defined.

The existing theorem instead depends logarithmically on every consecutive gap and a leading spectral condition number. The proposed bound would explain why a block can accommodate clusters of up to b singular values. It does not ask for a finite-precision extension.

References

1. T. Chen, E. N. Epperly, R. A. Meyer, C. Musco, and A. Rao, *Does block size matter in randomized block Krylov low-rank approximation?*, arXiv:2508.06486v2. Section 5 states the conjecture; Section 3.3 gives the full quantitative Theorem 1.3, and Algorithm 1 fixes the algorithm. Paper.
2. R. A. Meyer, C. Musco, and C. Musco, *On the Unreasonable Effectiveness of Single Vector Krylov Methods for Low-Rank Approximation*, SODA 2024, pp. 811–845, especially Theorem 4.5. Paper.

Status check — 2026-09-08

Searched the exact 2025 paper title, "block Krylov b-th order gap conjecture", and "randomized block Krylov clustered gaps 2026"; checked the current arXiv abstract/version record and Section 5. No proof or counterexample was found. Input perturbation results in Section 3.5 do not establish the unperturbed statement above.